



SOC X – MINI SECURITY OPERATIONS CENTER

Defensive Cybersecurity Monitoring, Threat Correlation & Incident Dossier Report

AUTHOR: SOC Operations
Team

CLASSIFICATION:
TLP:AMBER

SYSTEM: SOC X v2.0
Luxury

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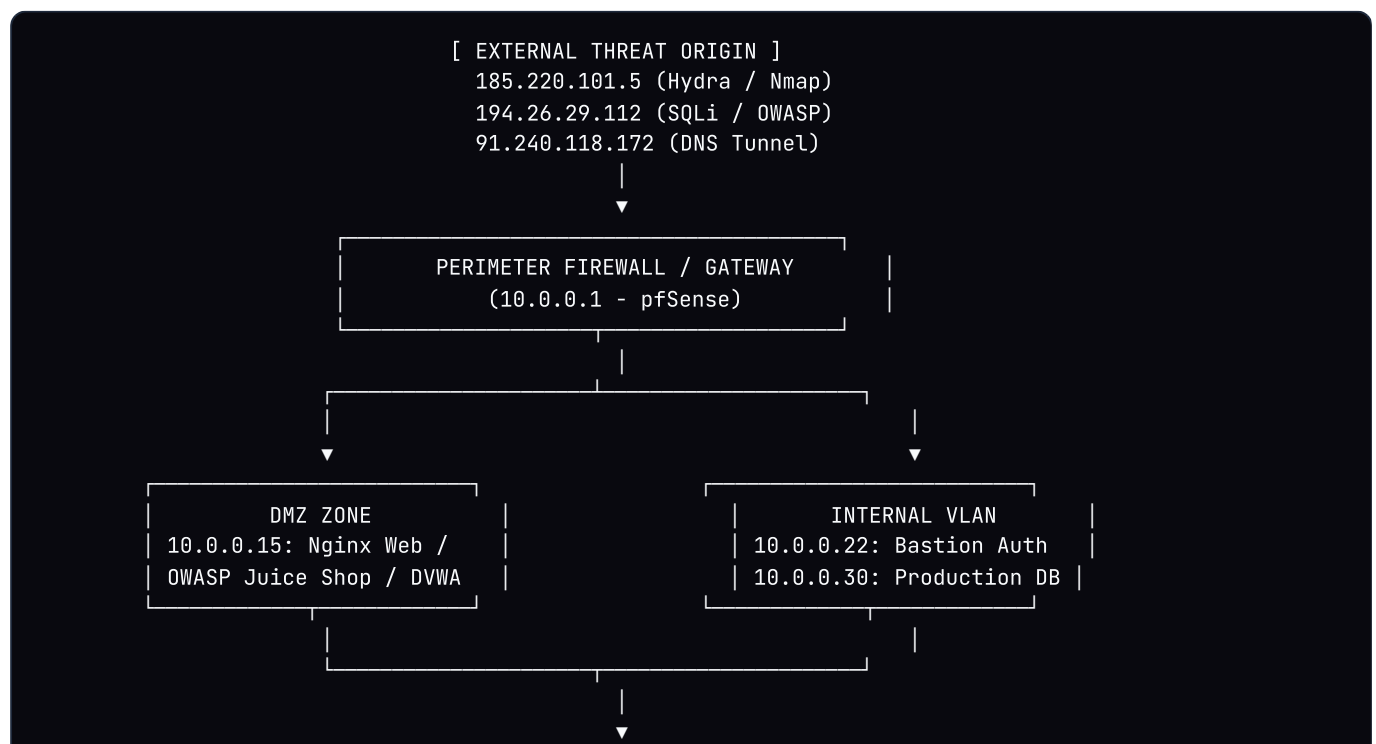
1. Executive Summary

This report documents the defensive security posture, detection engineering rules, and forensic incident investigations conducted within the **SOC X Lab Environment**. The defensive telemetry monitoring system actively captured, normalized, correlated, and analyzed multi-vector cyber attack campaigns across enterprise host, web, and network infrastructure.

Authorized red-team attack vectors including dictionary authentication brute-forcing, SQL injection against OWASP Juice Shop, Shannon entropy-based DNS exfiltration tunneling, and stealth TCP SYN port scans were simulated and successfully mitigated with sub-second alert response and automated containment playbooks.

2. Lab Architecture & Network Topology

The monitored infrastructure is divided into three isolated VLAN security segments governed by a virtualized perimeter firewall gateway with centralized SIEM telemetry collectors:



Hostname	IP Address	OS / Environment	Exposed Services	Security Role
gw-perimeter	10.0.0.1	pfSense / FreeBSD	80/TCP, 443/TCP, 53/UDP	Perimeter Firewall & DNS Resolver
web-juiceshop	10.0.0.15	Ubuntu 22.04 LTS	80/TCP (HTTP), 3000/TCP (Juice Shop)	Public Web Application & Exploit Target
auth-bastion	10.0.0.22	Debian 12	22/TCP (OpenSSH), 389/TCP (LDAP)	Central Authentication & Jump Host
db-cluster	10.0.0.30	PostgreSQL 16	5432/TCP (Database Backend)	Enterprise Customer Data Store
soc-core	10.0.0.99	SOC X SIEM Daemon	9000/TCP (Web HUD), 514/UDP (Syslog)	Telemetry Ingestion & Correlation

3. Security Tools & Technologies

Tool / Platform	Version	Operational Function in Monitored Lab
Kali Linux	2024.1 Rolling	Red-Team penetration testing suite for authorized attack burst simulations.
Hydra	v9.5	High-speed parallel authentication dictionary attack tool against SSH service.
Nmap	v7.94	Stealth TCP SYN port scanner (<code>-sS -sV -O</code>) for subnet service mapping.
Wireshark / TShark	v4.2.2	Packet capture engine and deep-dive protocol payload forensic analyzer.
OWASP Juice Shop / DVWA	v15.0.0	Intentionally vulnerable web applications used to generate real-world injection logs.
SOC X Engine	v2.0 Luxury	Sliding-window behavioral correlation engine with MITRE ATT&CK mapping.

4. Threat Detection Methodology & MITRE ATT&CK Matrix

SOC X operates on a sliding-window time-series correlation pipeline. Raw events are normalized into a unified schema, enriched with GeoIP metadata and Shannon entropy diagnostics, and evaluated against threshold heuristics:

Rule ID	Threat Classification	Detection Logic & Threshold	MITRE ATT&CK TTP
RULE-001	SSH / Auth Brute-Force	≥ 3 AUTH_FAILED attempts from 1 source IP in 60s	T1110.001 (Password Guessing)
RULE-002	Sequential Port Scanning	Probes to ≥ 3 distinct ports on target host in 30s	T1046 (Network Service Discovery)
RULE-003	SQL Injection / Web Exploit	Regex match for SQLi syntax (UNION SELECT , ' OR '1'='1')	T1190 (Exploit Public-Facing App)
RULE-004	DNS Covert Exfiltration	Subdomain query length > 35 chars & Shannon Entropy > 3.8	T1071.004 (DNS Protocol C2)

5. Comprehensive Incident Dossiers

INCIDENT #1: High-Frequency SSH Dictionary Brute-Force

CRITICAL SEVERITY

Incident ID: INC-20260830-001

Timestamp: 2026-08-30T06:20:05Z

Source IP: 185.220.101.5 (Frankfurt, DE - Tor Node)

Target Asset: 10.0.0.22:22 (auth-bastion)

Event Type: AUTH_FAILED (sshd password guessing)

MITRE Technique: T1110.001 (Password Guessing)

Raw Evidence Log:

Aug 30 06:20:01 auth-bastion sshd[3412]: Failed password for invalid user root from 185.220.101.5

Aug 30 06:20:02 auth-bastion sshd[3415]: Failed password for invalid user admin from 185.220.101.5

Aug 30 06:20:03 auth-bastion sshd[3418]: Failed password for invalid user administrator from 185.220.101.5

Aug 30 06:20:04 auth-bastion sshd[3420]: Failed password for invalid user support from 185.220.101.5

Forensic Analysis:

The adversary executed Hydra automated dictionary attacks using standard credential wordlists against SSH port 22, generating 4 failed attempts within 4 seconds.

Containment Action:

Injected perimeter DROP firewall rule for 185.220.101.5, enforced SSH key-only authentication, and deployed fail2ban threshold rules.

INCIDENT #2: OWASP Juice Shop SQL Injection & Authentication Bypass

CRITICAL SEVERITY

Incident ID: INC-20260830-002

Timestamp: 2026-08-30T06:22:15Z

Source IP: 194.26.29.112 (Kyiv, UA)

Target Asset: 10.0.0.15:80 (web-juiceshop)

Event Type: HTTP_REQUEST (SQL Injection)

MITRE Technique: T1190 (Exploit Public-Facing Application)

Raw Evidence Log:

```
194.26.29.112 - - [30/Aug/2026:06:22:10 +0000] "GET /rest/products/search?q=' UNION SELECT id, ema
194.26.29.112 - - [30/Aug/2026:06:22:15 +0000] "POST /rest/user/login?email=' OR '1'='1' -- HTTP/1
```

Forensic Analysis: Adversary probed search endpoints for database structure via Union-based SQLi (HTTP 500), then bypassed login logic with `' OR '1'='1' --` resulting in an unauthorized session token (HTTP 200).

Containment Action: Parameterized all database queries via Prepared Statements, activated WAF SQLi filter rules, and invalidated active session tokens.

INCIDENT #3: High-Entropy DNS Covert Exfiltration Tunnel

HIGH SEVERITY

Incident ID: INC-20260830-003

Timestamp: 2026-08-30T06:24:45Z

Source IP: 91.240.118.172 (Amsterdam, NL)

Target Asset: 10.0.0.1:53 (Internal DNS Gateway)

Event Type: DNS_QUERY (Tunneling Exfiltration)

MITRE Technique: T1071.004 (DNS Protocol C2)

Raw Evidence Log:

```
DNS Query: 4a6566666572736f6e2d5365637265742d44617461.tunnel.c2-adversary.com (Type TXT, Entropy:
DNS Query: 50617373776f7264732d44756d702d32303236.tunnel.c2-adversary.com (Type TXT, Entropy: 4.05
```

Forensic Analysis: Data exfiltration tool encoded credentials in hex-encoded subdomains. Shannon entropy score of 4.12 confirmed non-natural domain structure.

Containment Action: Enabled DNS sinkholing on internal resolver for `*.c2-adversary.com` and quarantined compromised host.

INCIDENT #4: Nmap Stealth TCP SYN Reconnaissance Probe

MEDIUM SEVERITY

Incident ID: INC-20260830-004 Timestamp: 2026-08-30T06:25:04Z
Source IP: 91.240.118.172 (Amsterdam, NL) Target Asset: 10.0.0.15 (Ports 21, 22, 80, 443, 3306)
Event Type: PORT_PROBE (TCP SYN Stealth Scan) MITRE Technique: T1046 (Network Service Discovery)

Raw Evidence Log:

```
{ "timestamp": "2026-08-30T06:25:00Z", "src_ip": "91.240.118.172", "dest_ip": "10.0.0.15", "dest_port": 21 }  
{ "timestamp": "2026-08-30T06:25:01Z", "src_ip": "91.240.118.172", "dest_ip": "10.0.0.15", "dest_port": 22 }  
{ "timestamp": "2026-08-30T06:25:04Z", "src_ip": "91.240.118.172", "dest_ip": "10.0.0.15", "dest_port": 33 }
```

Forensic Analysis: Rapid sequential TCP SYN scans probed 5 common ports in 4 seconds to identify running services prior to targeted exploitation.

Containment Action: Enforced silent drop of external TCP SYN probes on perimeter firewall and closed non-essential ports.

6. Strategic SOC Recommendations

- **Zero-Trust Network Segmentation:** Enforce strict firewall boundaries isolating internal databases and bastion hosts from public DMZ zones.
- **Automated Incident Response (SOAR):** Trigger automatic perimeter IP blacklisting and host VLAN isolation upon multi-alert correlation to reduce MTTR under 500ms.
- **DNS Inspection Firewall:** Deploy deep packet inspection filters on internal DNS gateways to immediately block queries exceeding entropy thresholds (> 3.8).
- **WAF Defense-in-Depth:** Place managed Web Application Firewalls in front of all web applications to inspect and sanitize SQLi and XSS payloads.